



Malicious Bot Detection on Twitter Using Reinforcement Learning and URL Feature Analysis

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Abstract



- Malicious social bots on Twitter spread spam, phishing links, and misinformation.
- The project uses Learning Automata (LA) to detect bots by learning from past decisions.
- URL features such as domain reputation, redirection count, and link frequency help identify suspicious accounts.
- LA updates detection decisions based on rewards for correct classifications and penalties for errors.
- The system stores tweet and URL data in a database for analysis.

Introduction



- Twitter has become a major platform for real-time communication, but its openness also attracts malicious bots that spread spam, phishing URLs, and misleading information.
- This project focuses on detecting malicious bots on Twitter using Reinforcement Learning (RL) combined with URL feature analysis.
- URL-based features such as link frequency, domain age, redirection count, and blacklist status are analyzed to identify suspicious behavior.
- The system is implemented using Java for application logic, MySQL for storing user data, tweets, and extracted features, and Apache Tomcat as the web server for deploying the application.

Literature Survey



- **Nithin et al. (2025)** Developed a reinforcement learning model using URL feature analysis to effectively identify malicious bots on Twitter.
- **Liu et al. (2023)** Proposed OFA, a graph-based model with frequency-adaptive attention to accurately detect bots hidden among real users.
- **Yang et al. (2022)** Introduced RoSGAS, a reinforcement learning–based graph neural network that adapts bot detection using social network .
- **Rout et al. (2020)** Proposed a Learning Automata model using URL features like redirection for spam and bot detection, improving accuracy through adaptive learning.

Existing Method



- Traditional methods for detecting malicious bots on Twitter primarily use machine learning and rule-based approaches.
- They rely on features like tweet frequency, follower-to-following ratios, profile age, and content patterns, including hashtags, mentions, and repeated messages.
- Some methods also analyze URLs based on domain reputation, link shortening, and sharing frequency to identify suspicious activity.
- They also face challenges in handling the large volume of Twitter data in real time, leading to reduced accuracy and scalability.

Disadvantages



- **Slow Response Time:** Many existing approaches are not designed for real-time detection, causing delays in identifying malicious activity.
- **Manual Updates Required:** Rule-based approaches need frequent manual intervention to update detection rules and making them time-consuming.
- **Scalability Issues:** Handling large volumes of Twitter data in real-time can be challenging, leading to delays in detection.
- **Static Learning:** Most traditional models do not improve over time, meaning their detection performance may degrade as bots evolve.

Proposed Method



- The proposed method detects malicious bots on Twitter using Reinforcement Learning (RL) combined with URL feature analysis. Key URL features such as domain reputation, redirection patterns, link frequency, and blacklisting are extracted and stored in MySQL.
- The RL model learns to classify accounts as bots or genuine users by receiving rewards for correct predictions and penalties for misclassifications, improving detection accuracy over time.
- The system is implemented in Java for application logic and deployed on Apache Tomcat for real-time monitoring.

Advantages



- **Real-Time Detection:** The method supports continuous monitoring of Twitter activity for immediate identification of malicious bots.
- **URL-Based Precision:** Analysis of URL features like domain reputation, redirections, and link frequency enhances detection of sophisticated spam and phishing bots.
- **Scalable Implementation:** Using Java, MySQL, and Tomcat ensures the system can handle large datasets efficiently.
- **Automated Decision-Making:** Minimal manual intervention is required, reducing maintenance and improving efficiency.

Tools



Java

Purpose: Java is used to develop the core logic of the system, including reinforcement learning algorithms and URL feature extraction for bot detection.

Features: Platform independent, secure, object-oriented, supports multithreading, and has rich libraries.

Example: Java analyzes tweets, extracts URL features, applies reinforcement learning, and classifies Twitter accounts as malicious or genuine.



MySQL

Purpose: MySQL is used to store and manage Twitter user data, tweets, URLs, and extracted features.

Features: Relational database, fast SQL queries, data integrity, scalability, and easy integration with Java.

Apache Tomcat

Purpose: Apache Tomcat is used to deploy and run the Java-based web application for malicious bot detection. It handles client requests and enables real-time monitoring.

Features: Tomcat supports Servlets and JSP, provides secure request handling, ensures scalability, and is lightweight and open-source.

Hardware Requirements



H/W System Configuration:-

Processor	- Pentium –IV
RAM	- 4 GB (min)
Hard Disk	- 20 GB
Key Board	- Standard Windows Keyboard
Mouse	- Two or Three Button Mouse
Monitor	- SVGA

Software Requirements



Operating System - Windows XP

Coding Language - Java/J2EE(JSP,Servlet)

Front End - J2EE

Back End - MySQL

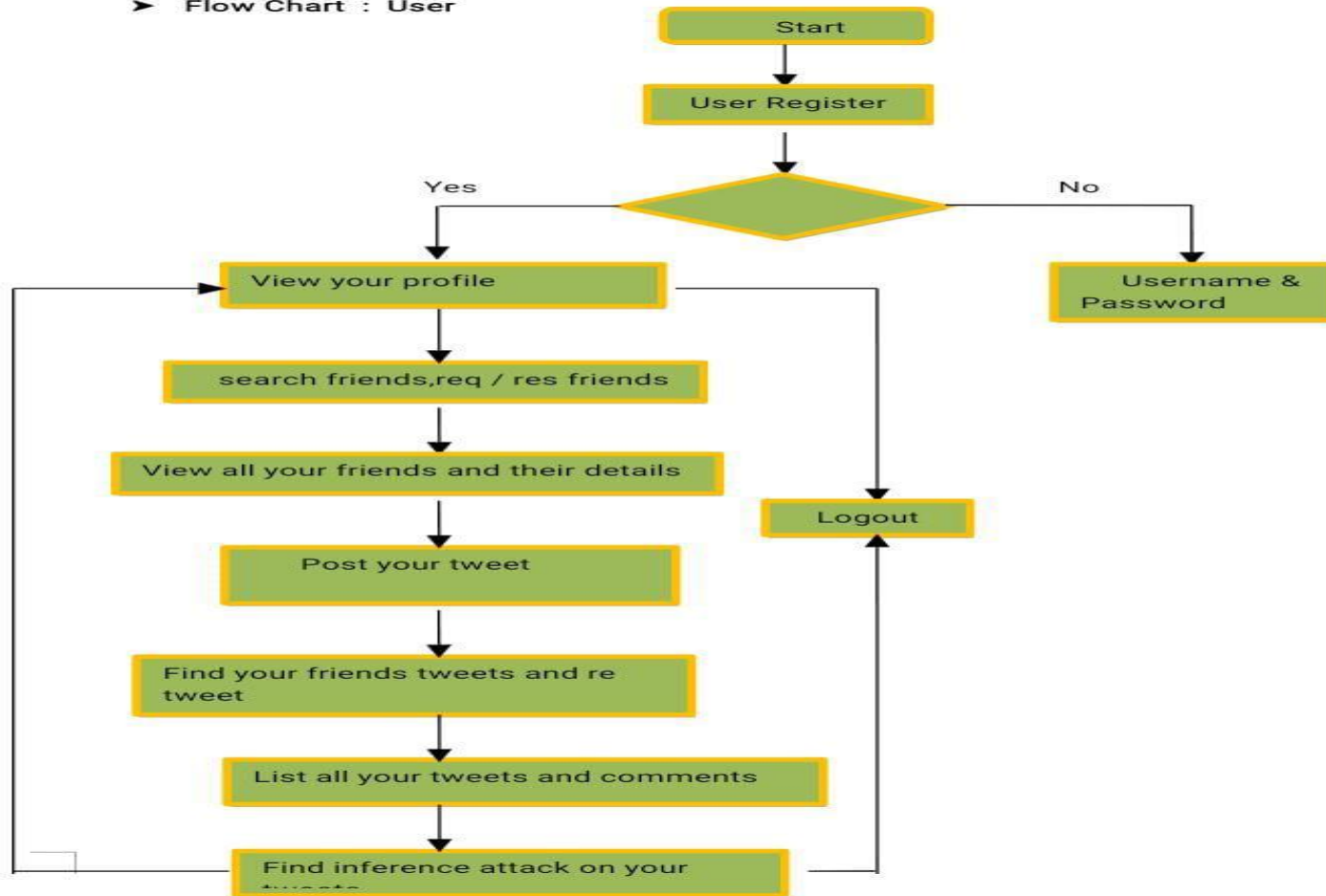
Methodologies



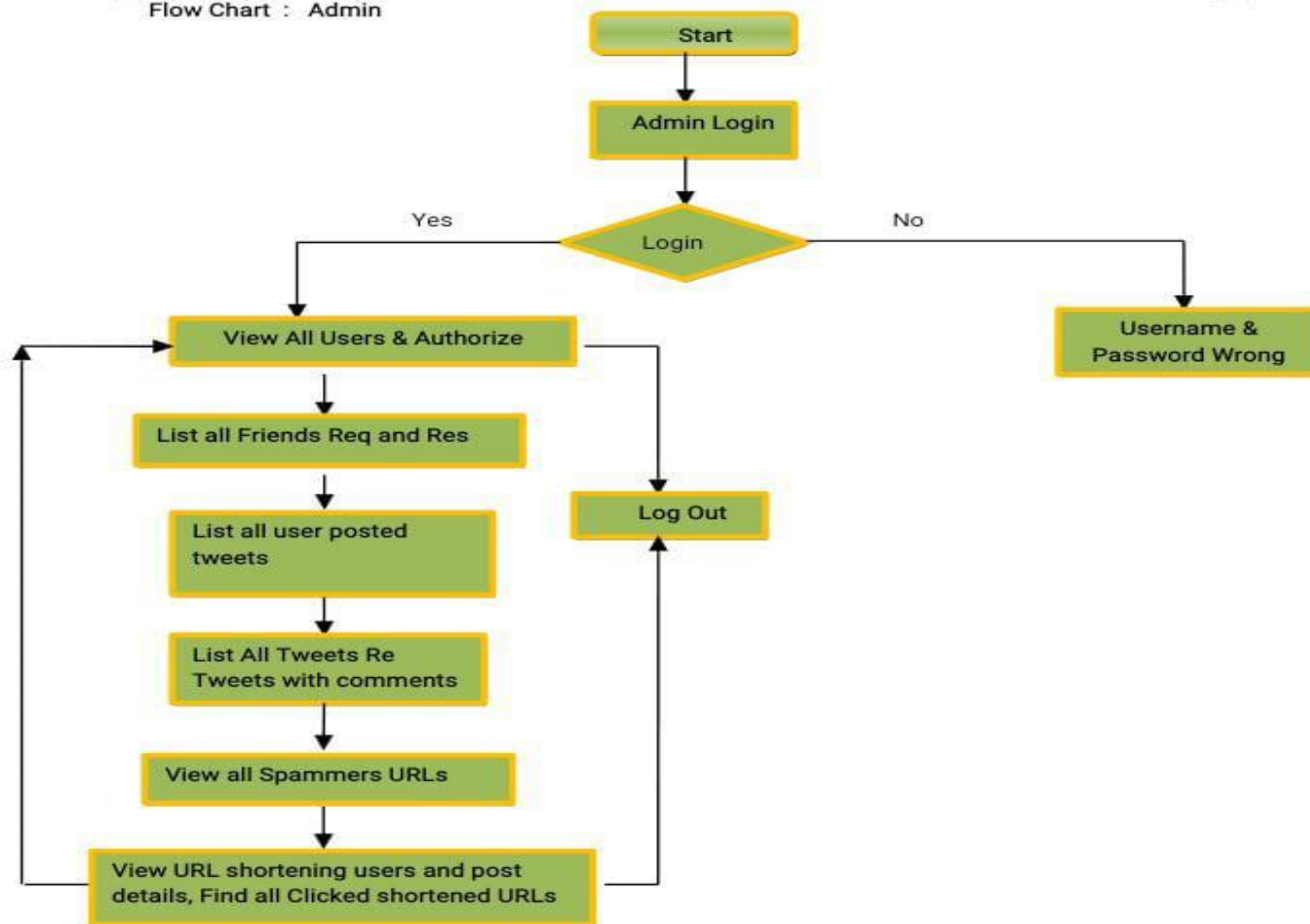
- **Data Collection:** Gather Twitter user data, tweets, and URLs.
- **Data Preprocessing:** Remove duplicates, inactive users, and invalid URLs.
- **URL Feature Extraction:** Extract domain age, URL frequency, redirection count, and blacklist status.
- **State & Action Definition:** Use features as states; classify accounts as bot or genuine.
- **Reinforcement Learning Training:** Train model with rewards for correct and penalties for wrong detection.
- **Bot Classification:** Detect malicious bots in real time.
- **Data Storage:** Save features and results in MySQL via JDBC.
- **Deployment:** Run system on Apache Tomcat with Java and display results on a web interface.

Flow diagrams

➤ Flow Chart : User

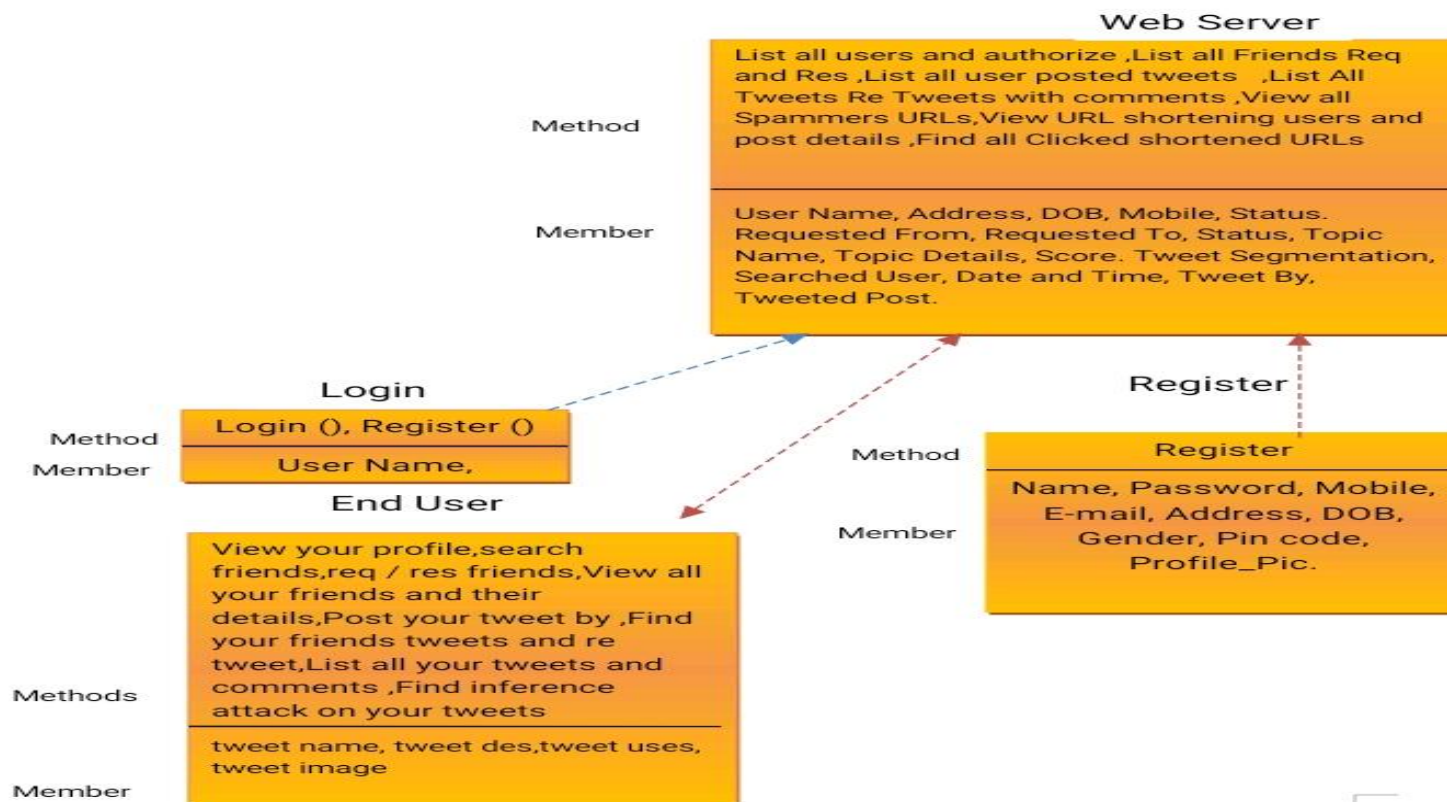


Flow Chart : Admin



Class diagrams

Class Diagram



Result & Discussion



The image displays the Apache Tomcat 7.0 server startup log, showing initialization and deployment processes

It confirms the successful deployment of a web application for identity theft detection and that the server is running without errors.

```
Select C:\Program Files\Apache Software Foundation\Tomcat 7.0\bin\Tomcat7.exe
Feb 03, 2026 10:44:35 AM org.apache.catalina.core.AprLifecycleListener init
INFO: The APR based Apache Tomcat Native library which allows optimal performance in production environments was not found on the java.library.path: C:\Program Files\Apache Software Foundation\Tomcat 7.0\bin;C:\WINDOWS\Sun\Java\bin;C:\WINDOWS\system32;C:\WINDOWS;C:\ProgramData\Oracle\Java\javapath;C:\Program Files (x86)\VMware\VMware Workstation\bin;C:\Program Files\Common Files\Oracle\Java\javapath;C:\Users\KEERTHI GARA\AppData\Local\Programs\Python\Python313;C:\Users\KEERTHI GARA\AppData\Local\Programs\Python\Python37\Scripts;C:\Users\KEERTHI GARA\AppData\Local\Programs\Python\Python37;C:\Users\KEERTHI GARA\AppData\Local\Microsoft\WindowsApps;C:\Users\KEERTHI GARA\AppData\Local\Programs\Microsoft VS Code\bin;C:\Program Files\MySQL\MySQL Server 8.0\bin;C:\Program Files\Java\jdk-21\bin;C:\Program Files\Java\jdk1.8.0_161\bin;.
Feb 03, 2026 10:44:35 AM org.apache.coyote.AbstractProtocol init
INFO: Initializing ProtocolHandler ["http-bio-8080"]
Feb 03, 2026 10:44:35 AM org.apache.coyote.AbstractProtocol init
INFO: Initializing ProtocolHandler ["ajp-bio-8009"]
Feb 03, 2026 10:44:35 AM org.apache.catalina.startup.Catalina load
INFO: Initialization processed in 496 ms
Feb 03, 2026 10:44:35 AM org.apache.catalina.core.StandardService startInternal
INFO: Starting service Catalina
Feb 03, 2026 10:44:35 AM org.apache.catalina.core.StandardEngine startInternal
INFO: Starting Servlet Engine: Apache Tomcat/7.0.42
Feb 03, 2026 10:44:35 AM org.apache.catalina.startup.HostConfig deployDirectory
INFO: Deploying web application directory C:\Program Files\Apache Software Foundation\Tomcat 7.0\webapps\Detection of Malicious Social Bots Using Learning Automata With URL Features in Twitter Network
Feb 03, 2026 10:44:35 AM org.apache.catalina.loader.WebappClassLoader validateJarFile
INFO: validateJarFile(C:\Program Files\Apache Software Foundation\Tomcat 7.0\webapps\Detection of Malicious Social Bots Using Learning Automata With URL Features in Twitter Network\WEB-INF\lib\servlet-api.jar) - jar not loaded. See Servlet Spec 2.3, section 9.7.2. Offending class: javax/servlet/Servlet.class
Feb 03, 2026 10:44:36 AM org.apache.catalina.util.SessionIdGenerator createSecureRandom
INFO: Creation of SecureRandom instance for session ID generation using [SHA1PRNG] took [401] milliseconds.
Feb 03, 2026 10:44:36 AM org.apache.catalina.startup.HostConfig deployDirectory
INFO: Deploying web application directory C:\Program Files\Apache Software Foundation\Tomcat 7.0\webapps\docs
Feb 03, 2026 10:44:36 AM org.apache.catalina.startup.HostConfig deployDirectory
INFO: Deploying web application directory C:\Program Files\Apache Software Foundation\Tomcat 7.0\webapps\manager
Feb 03, 2026 10:44:36 AM org.apache.catalina.startup.HostConfig deployDirectory
INFO: Deploying web application directory C:\Program Files\Apache Software Foundation\Tomcat 7.0\webapps\ROOT
Feb 03, 2026 10:44:36 AM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["http-bio-8080"]
Feb 03, 2026 10:44:36 AM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["ajp-bio-8009"]
Feb 03, 2026 10:44:36 AM org.apache.catalina.startup.Catalina start
INFO: Server startup in 1214 ms
```

Result & Discussion



First, we need to open Tomcat. Then, search for "localhost" on Google. Go to the Twitter server and click.

The screenshot shows a web browser window with the address bar displaying `localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/`. The page title is "Detection of Malicious Social Bots Using Learning Automata With URL Features in Twitter Network". The main content area features a large graphic titled "Twitter - Online Social Network" with a network of user avatars. Below this, there is a section titled "Twitter - Detection of Malicious Social Bots" which includes a diagram and a detailed text description of the malicious social bot detection process. A sidebar menu on the right lists "Home", "Users", and "Tweet Server".

Twitter - Detection of Malicious Social Bots

Malicious social bots generate fake tweets and automate their social relationships either by pretending like a follower or by creating multiple fake accounts with malicious activities. Moreover, malicious social bots post shortened malicious URLs in the tweet in order to redirect the requests of online social networking participants to some malicious servers. Hence, distinguishing malicious social bots from legitimate users is one of the most important tasks in the Twitter network. To detect malicious social bots, extracting URL-based features (such as URL redirection, frequency of shared URLs, and spam content in URL) consumes less amount of time in comparison with social graph-based features (which rely on the social interactions of users). Furthermore, malicious social bots cannot easily manipulate URL redirection chains. In this article, a learning automata-based malicious social bot detection (LA-MSBD) algorithm is proposed by integrating a trust computation model with URL-based features for identifying trustworthy participants (users) in the Twitter network. The proposed trust computation model contains two parameters, namely, direct trust and indirect trust. Moreover, the direct trust is derived from Bayes' theorem, and the indirect trust is

Twitter - Online Social Network

Twitter is a popular online social network service for sharing short messages (tweets) among friends.

Sidebar Menu

- Home
- Users
- Tweet Server

Concepts

Learning automata (LA),
malicious social bots,
online social networks (OSNs),
trust.

Result & Discussion



You need to log in by providing the server name and password.

Home User Tweet Server

Twitter - Online Social Network

Twitter is a popular online social network service for sharing short messages (tweets) among friends.

Welcome to Tweet Server Login Page..

Server Name (required)

Password (required)

Back

Sidebar Menu

[Home](#)

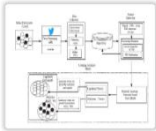
[Index Page](#)

Result & Discussion



After logging in, a page like this will open. Ahh, next, go to View and Authorized Users. Next, go back to the page. Next, go to Logout, right-click on it, and then click on 'Open in new tab'..

Welcome to Tweet Server Page ..



Malicious social bots generate fake tweets and automate their social relationships either by pretending like a follower or by creating multiple fake accounts with malicious activities. Moreover, malicious social bots post shortened malicious URLs in the tweet in order to redirect the requests of online social networking participants to some malicious servers. Hence, distinguishing malicious social bots from legitimate users is one of the most important tasks in the Twitter network. To detect malicious social bots, extracting URL-based features (such as URL redirection, frequency of shared URLs, and spam content in URL) consumes less amount of time in comparison with social graph-based features (which rely on the social interactions of users). Furthermore, malicious social bots cannot easily manipulate URL redirection chains. In this article, a learning automata-based malicious social bot detection (LA-MSBD) algorithm is proposed by integrating a trust computation model with URL-based features for identifying trustworthy participants (users) in the Twitter network. The proposed trust computation model contains two parameters, namely, direct trust and indirect trust. Moreover, the direct trust is derived from Bayesâ€™ theorem, and the indirect trust is derived from the Dempsterâ€™ Shafer theory (DST) to determine the trustworthiness of each participant accurately. Experimentation has been performed on two Twitter data sets, and the results illustrate that the proposed algorithm achieves improvement in precision, recall, F-measure, and accuracy compared with existing approaches for MSBD.

Server Menu

- [Home](#)
- [View and Authorize Users](#)
- [View/Add Short URLs](#)
- [View All User Posted Tweets](#)
- [View All Spammer URLs](#)
- [View Friend Request/Response](#)
- [View All Re-Tweets and Comments](#)
- [View All URL Shortening Bots Users](#)
- [View NO.of time Posted URL Shortening Bots Users](#)
- [View Clicked URLs Shortened Bots Users](#)
- [View NO.of Time URL Shortening Bots Users](#)
- [Log Out](#)

Result & Discussion



Click on Register

Browser tabs: Tweet Server Main Page.., User Login Page..

Address bar: localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/UserLogin.jsp

Detection of Malicious Social Bots Using Learning Automata With URL Features in Twitter Network

Home User Tweet Server



Twitter Online Social Network

Twitter is a popular online social network service for sharing short messages (tweets) among friends.

Welcome To User Login..



Name (required)

Password (required)

[New User?](#) [Register](#)

Sidebar Menu

- [Home](#)
- [Index Page](#)

[Back](#)

Result & Discussion



The registration page will open like that. Enter your details and fill out the form, then finally click on Register.

Twitter Server Main Page.. x User Registration Page.. +

localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/UserRegister.jsp

Home

User Login

Welcome to User Registration..

User Name (required)
keerthi

Password (required)

Email Address (required)
22kq1a4707iot@gmail.com

Mobile Number (required)
6304573614

Your Address
Kandukur

Date of Birth (required)
15-06-2005

Select Gender (required)
FEMALE

Select Profile Picture (required) Choose File User1.jpg

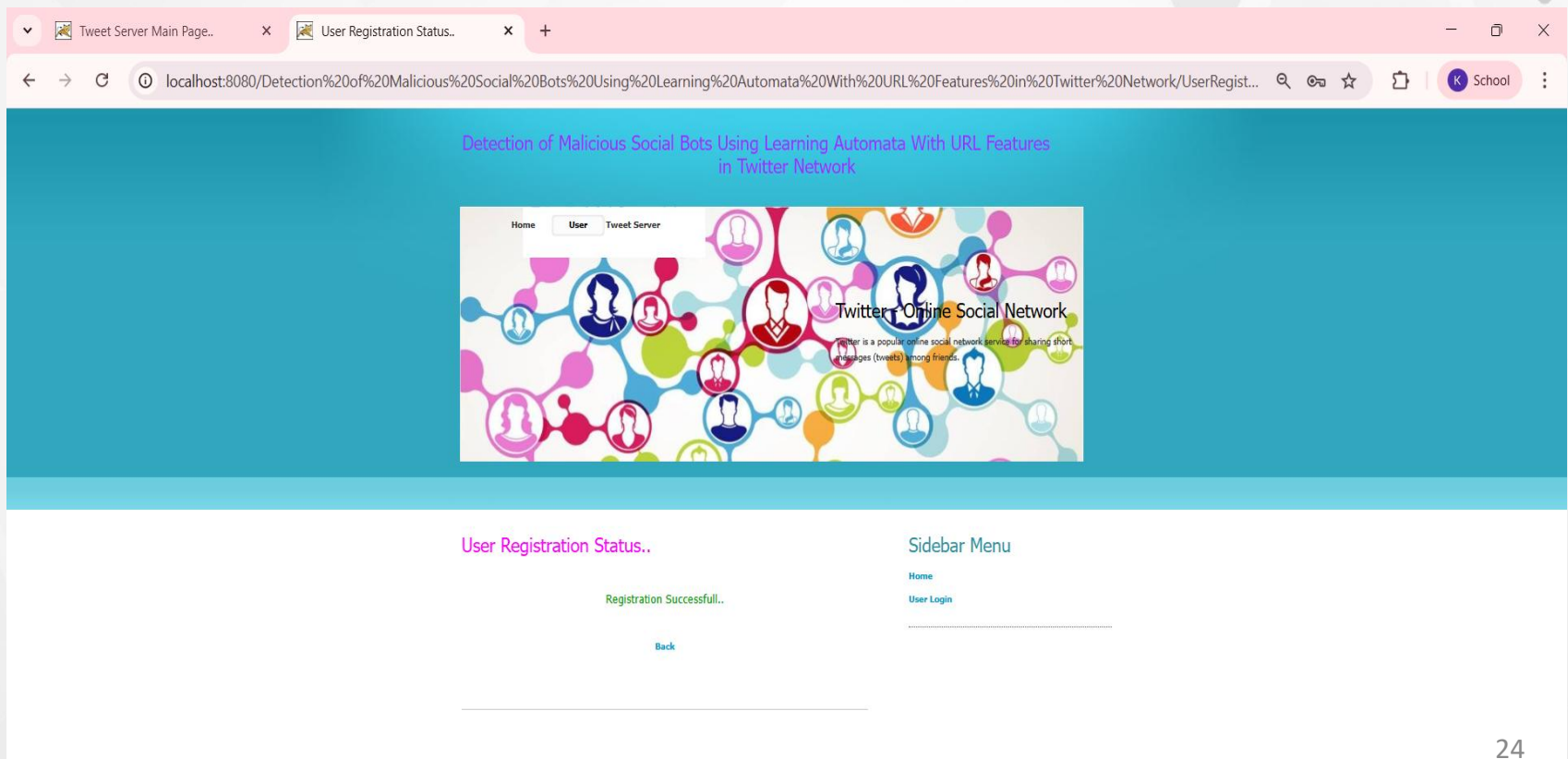
REGISTER

Back

Result & Discussion



After filling out the registration form, you will receive a message saying "Registration successful".



Result & Discussion



Register with the same process but with a different name.

Twitter Server Main Page.. x User Registration Page.. +

localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/UserRegister.jsp

Welcome to User Registration..

User Name (required)
asadhi

Password (required)

Email Address (required)
22kq1a4717iot@gmail.com

Mobile Number (required)
7396443029

Your Address
Kandukur

Date of Birth (required)
27-10-2004

Select Gender (required)
MALE

Select Profile Picture (required) Choose File images (4).png

REGISTER

Back

Sidebar Menu
Home
User Login

Result & Discussion

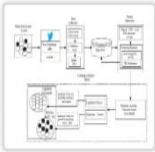


Next, go to Search Friends and search for the name.

messages (tweets) among friends.

Result & Discussion

Welcome User keerthi ..



Malicious social bots generate fake tweets and automate their social relationships either by pretending like a follower or by creating multiple fake accounts with malicious activities. Moreover, malicious social bots post shortened malicious URLs in the tweet in order to redirect the requests of online social networking participants to some malicious servers. Hence, distinguishing malicious social bots from legitimate users is one of the most important tasks in the Twitter network. To detect malicious social bots, extracting URL-based features (such as URL redirection, frequency of shared URLs, and spam content in URL) consumes less amount of time in comparison with social graph-based features (which rely on the social interactions of users). Furthermore, malicious social bots cannot easily manipulate URL redirection chains. In this article, a learning automata-based malicious social bot detection (LA-MSBD) algorithm is proposed by integrating a trust computation model with URL-based features for identifying trustworthy participants (users) in the Twitter network. The proposed trust computation model contains two parameters, namely, direct trust and indirect trust. Moreover, the direct trust is derived from Bayes's theorem, and the indirect trust is derived from the Dempster-Shafer theory (DST) to determine the trustworthiness of each

User Menu

- Home
- My Profile
- Search Friends**
- Create Tweets
- View My Friends
- View Friend Requests
- Search Tweets and Comment
- View My Tweets and Comments
- View Shortening Bots on My Tweets
- View Friends Tweets and Re-Tweet
- Log Out

Result & Discussion



After you perform the name search, a page will open like this, click on the request button.

localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/SearchFriends.jsp

Search Friends..

Enter Friend Name :

Search

Back

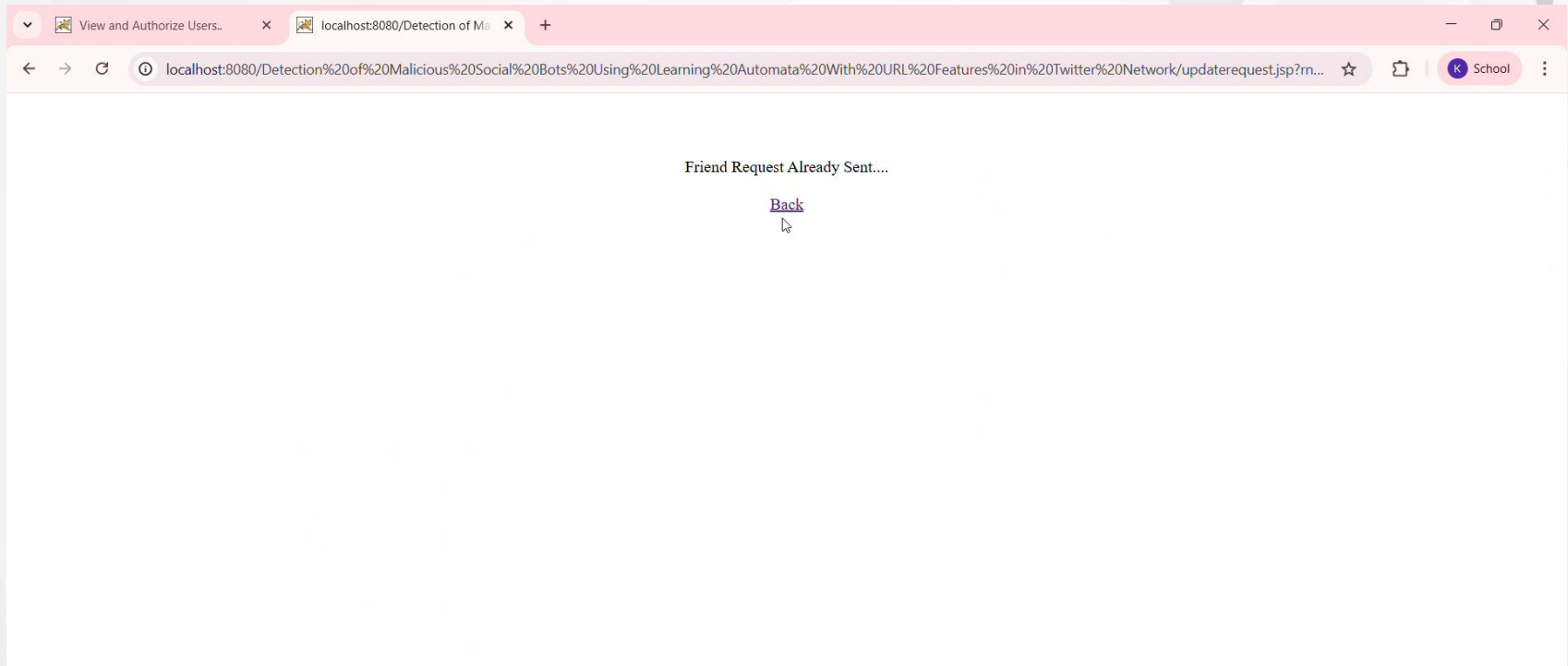
Results Found.. ..

	User Name :	aadhi
	E-Mail :	22kq1a4717lot@gmail.com
	Mobile :	7396443029
	Address :	Kandukur
	DOB :	27-10-2004
	Gender :	MALE
	Kind Of User :	Normal
Status :	Authorized	Request

Result & Discussion



The request will be sent, it will open like that. Next, go back and log out from there.



Result & Discussion



Next, go to the users' second registration page, enter the name and password, and log in.

View and Authorize Users.. x User Login Page.. x +

localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/UserLogin.jsp

Twitter - Online Social Network

Twitter is a popular online social network service for sharing short messages (tweets) among friends.

Welcome To User Login..

Name (required)

Password (required)

[New User?](#) [Register](#)

Sidebar Menu

[Home](#)

[Index Page](#)

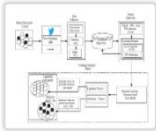
[Back](#)

Result & Discussion



After logging in, go to 'View My Friends', then go back, and then go to 'View Friend Requests'.

Welcome User aadhi ..



Malicious social bots generate fake tweets and automate their social relationships either by pretending like a follower or by creating multiple fake accounts with malicious activities. Moreover, malicious social bots post shortened malicious URLs in the tweet in order to redirect the requests of online social networking participants to some malicious servers. Hence, distinguishing malicious social bots from legitimate users is one of the most important tasks in the Twitter network. To detect malicious social bots, extracting URL-based features (such as URL redirection, frequency of shared URLs, and spam content in URL) consumes less amount of time in comparison with social graph-based features (which rely on the social interactions of users). Furthermore, malicious social bots cannot easily manipulate URL redirection chains. In this article, a learning automata-based malicious social bot detection (LA-MSBD) algorithm is proposed by integrating a trust computation model with URL-based features for identifying trustworthy participants (users) in the Twitter network. The proposed trust computation model contains two parameters, namely, direct trust and indirect trust. Moreover, the direct trust is derived from Bayes's theorem, and the indirect trust is derived from the Dempster-Shafer theory (DST) to determine the trustworthiness of each participant accurately. Experimentation has been performed on two Twitter data sets, and the results illustrate that the proposed algorithm achieves improvement in precision, recall, F-measure, and accuracy compared with existing approaches for MSBD.

User Menu

- Home
- My Profile
- Search Friends
- Create Tweets
- View My Friends**
- View Friend Requests
- Search Tweets and Comment
- View My Tweets and Comments
- View Shortening Bots on My Tweets
- View Friends Tweets and Re-Tweet
- Log Out

Result & Discussion



Click on Waiting, then go back to the list and click on View My Friends.

Friend Requests To aadhi..

Username	E-mail	Mobile	Address	Gender	Status
keerthi	22kq1a4707iot@gmail.com	6304573614	Kandukur	FEMALE	waiting

[Back](#)

Result & Discussion



Log out completely and then log in by entering Keerthi's name and password.

View and Authorize Users.. x User Login Page.. x +

localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/UserLogin.jsp

Twitter - Online Social Network

Twitter is a popular online social network service for sharing short messages (tweets) among friends.

Welcome To User Login..

Name (required)

Password (required)

Login [New User? Register](#)

Back

Sidebar Menu

[Home](#)

[Index Page](#)

Result & Discussion



After logging in, the next page will be displayed like this. Go to 'Create Tweets'.

The screenshot shows a web browser window with two tabs: 'View and Authorize Users...' and 'User Main Page..'. The address bar displays the URL: `localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/UserMain.jsp`. The page header features a decorative banner with colorful icons of people connected by lines. The main content area is divided into two columns. The left column, titled 'Welcome User keerthi ..', contains a diagram of a social network and a paragraph of text. The right column, titled 'User Menu', lists several navigation options. The 'Create Tweets' link in the menu is highlighted with a green underline and a mouse cursor.

Welcome User keerthi ..

Malicious social bots generate fake tweets and automate their social relationships either by pretending like a follower or by creating multiple fake accounts with malicious activities. Moreover, malicious social bots post shortened malicious URLs in the tweet in order to redirect the requests of online social networking participants to some malicious servers. Hence, distinguishing malicious social bots from legitimate users is one of the most important tasks in the Twitter network. To detect malicious social bots, extracting URL-based features (such as URL redirection, frequency of shared URLs, and spam content in URL) consumes less amount of time in comparison with social graph-based features (which rely on the social interactions of users). Furthermore, malicious social bots cannot easily manipulate URL redirection chains. In this article, a learning automata-based malicious social bot detection (LA-MSBD) algorithm is proposed by integrating a trust computation model with URL-based features for identifying trustworthy participants (users) in the Twitter network. The proposed trust computation model contains two parameters, namely, direct trust and indirect trust. Moreover, the direct trust is derived from Bayes's theorem, and the indirect trust is derived from the Dempster-Shafer theory (DST) to determine the trustworthiness of each participant accurately. Experimentation has been performed on two Twitter data sets, and the results illustrate that the proposed algorithm achieves improvement in precision, recall, F-

User Menu

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- My Profile
- Search Friends
- Create Tweets
- View My Friends
- View Friend Requests
- Search Tweets and Comment
- View My Tweets and Comments
- View Shortening Bots on My Tweets
- View Friends Tweets and Re-Tweet
- Log Out

Result & Discussion



Enter the tweet name, description, and users, then select an image. Click on 'Post Tweet'. Next, after tweets posted successfully, next go back.

View and Authorize Users.. Posting Tweets..

localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/U_CreateTweet.jsp

4

Posting Tweets..

Tweet Name

Tweet Description

Tweet Uses

Select Tweet Image

Choose File bit.png

Post Tweet

Result & Discussion



Next, click on 'View my tweets and comments'. A page will open like this. Next, click on 'Back'.



All My Posted Tweets and It's Comments..

ID	Tweet Image	Tweet Name	Tweet Description	Tweet Uses	Date
4		ebuy	best ecommerce site	we can buy easily I	21/01/2026 10:09:11

Result & Discussion



Next, to go the add short URLs and add some URLs. After come to back shorten the view, click on Botsa on my tweets, and then go back and log out.

Add Short URLs..

Enter Short URL

Add

[Back](#)

Added Short URLs..

Sl No.	Short URL Name
1	goo.gl
2	bit.ly
3	t.co

null

Result & Discussion



Again, aadi log in, go to the search tweets and comments section, and click on 'View Details' next to 'Tweet Found'.

The screenshot shows a web browser window with two tabs: 'All User Posted Tweets..' and 'Searching Tweets..'. The address bar shows the URL: `localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/SearchTweets.jsp`. The page has a header with a colorful banner featuring various icons. Below the header, the main content area is divided into two sections:

- Search Tweets and Comment..**: This section contains a form with the label 'Enter Keyword Name :', a text input field, and a 'Search' button. Below the form is a 'Back' link.
- Sidebar Menu**: This section contains two links: 'Home' and 'Log Out'.

Below the search section, there is a section titled 'Tweets Found..'. It displays a single result with a red square icon containing a white 'b' logo. Below the icon, the text 'Tweet Name : ebuy' is shown, followed by a 'View Details' link.

Result & Discussion



URL Shortenings.. x Searched Tweet Details.. x +

localhost:8080/Detection%20of%20Malicious%20Social%20Bots%20Using%20Learning%20Automata%20With%20URL%20Features%20in%20Twitter%20Network/SearchTweets1.jsp?tn... School

Tweet ebuy's Details..

	Tweet By	keerthi
	Tweet Description	best ecommerce site
	Tweet Uses	we can buy easily
	Date	21/01/2026 10:09:11
Tweet on This Post	good site with url bit.ly we can buy fast Post	

Sidebar Menu

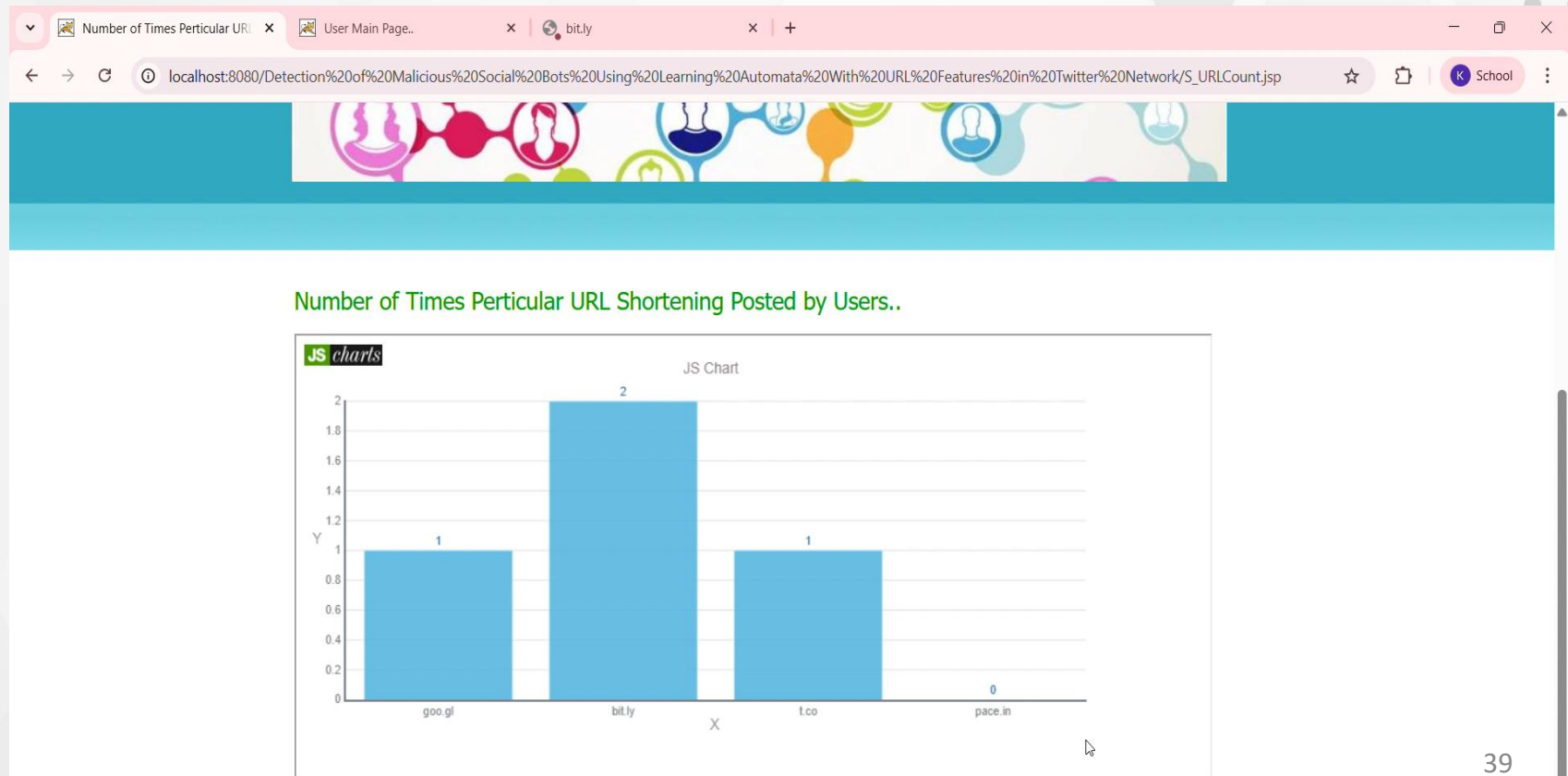
- Home
- Log Out

[Back](#)

Result & Discussion



Go to the server menu, view the number of times the shortened URLs have been clicked by users.



Conclusion



- This article introduces an LA-MSBD method to detect malicious bots by checking URL features and user trust in tweets.
- It uses learning techniques to gradually identify users who share harmful links.
- Tests on two Twitter datasets show better accuracy (up to 7%) than existing methods, with high precision.
- Future work will focus on how different features affect bot detection.

Future Scope



- Extend detection to other platforms like Instagram and Facebook
- Use deep learning for higher accuracy
- Detect bots in real time
- Analyze images, videos, and text along with URLs
- Improve scalability for large datasets

References



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Thank
you